RISK CLASSIFICATION

Immediate vs. Potential Risk

IMMEDIATE

VS.

POTENTIAL

AGENDA

01

Objective

Purpose and goals of this risk classification framework

02

Immediate Risk — Definition & Characteristics

What constitutes an immediate risk and when to act now

03

Potential Risk — Definition & Characteristics

Understanding trends, anomalies and early warning signals

04

Scenarios & Examples

7 real-world scenarios classified and explained

05

Quick Decision Framework

3-step triage process for fast, accurate classification

06

Summary & Key Takeaways

Comparison table and guiding principles

OBJECTIVE

This presentation establishes a clear, actionable framework for classifying operational and security alerts as either Immediate Risk or Potential Risk, enabling teams to triage faster, respond appropriately, and avoid the twin pitfalls of underreaction and overreaction.

Define clear thresholds

Establish consistent criteria that distinguish active, imminent threats from future risks

Guide triage decisions

Give responders a repeatable 3-step decision process during live incidents

Reduce MTTD & MTTR

Minimize Mean Time to Detect and Respond by removing classification ambiguity

Prevent escalation

Catch potential risks early before they become immediate incidents

IMMEDIATE RISK

Definition & Key Characteristics

Any condition detected by an alert that indicates active or imminent impact to production systems, users, security, or data, requiring urgent action within minutes to prevent or limit damage.

Timeframe

Impact happening NOW or within minutes

User Impact

Active outage, degradation, or SLA breach

Blast Radius

Multiple users, services, or regions affected

Urgency Signals

P1/P2 pages, high error rates, health check failures

Required Response: Respond immediately (0–5 min) · Begin mitigation before full diagnosis · Escalate if needed · Prioritize stabilization

POTENTIAL RISK

Definition & Key Characteristics

Any condition where an alert indicates a trend, anomaly, or vulnerability that could lead to future impact, but no active or imminent user-facing damage is occurring yet.

Timeframe

Impact hours, days away, or conditional on other events

Current Impact

No outage; system functioning within acceptable thresholds

Risk Level

Probabilistic requires progression or additional triggers

Reversibility

Time exists to plan, investigate, and prevent escalation

Required Response: Acknowledge within SLA (15–60 min) · Investigate & create mitigation plan · Track via ticket · Schedule remediation

Part 1: Scenarios 1–4

01 API Error Rate Spike

IMMEDIATE

Alert: Error rate jumps 0.5% → 35% across all API endpoints

Reason: Active user impact, large blast radius, likely ongoing outage

Response: Rollback, failover, or isolate faulty service

02 Gradual Memory Increase

POTENTIAL

Alert: Memory usage: 40% → 75% over 6 hours — possible leak

Reason: No current failure; trend suggests future exhaustion; time exists before impact

Response: Investigate within hours; plan restart, patch, or scaling

03 Database CPU at 100%

IMMEDIATE

Alert: Primary DB CPU maxed; query latency increased 10x

Reason: Active degradation, likely cascading failures, core dependency stressed

Response: Throttle traffic, kill expensive queries, scale or failover

04 Disk Usage at 85%

POTENTIAL

Alert: Disk steadily increasing; projected 100% in ~12 hours

Reason: No current outage; predictable future failure; sufficient remediation window

Response: Same-day: clean up logs, increase storage, adjust retention

Part 2: Scenarios 5–7

05 Multiple Failed Login Attempts

POTENTIAL*

Alert: 500 failed login attempts over 10 min from single IP; no successful breaches

Reason: No confirmed compromise yet; could escalate to brute-force; security signal not impact **Response:** Block IP, enable rate limiting, monitor for escalation

06 Health Check Failing — One Region

IMMEDIATE

Alert: Health checks failing for service in one region; traffic still routed globally

Reason: Partial outage occurring; risk of traffic spillover; redundancy being consumed **Response:** Remove region from rotation, investigate root cause immediately

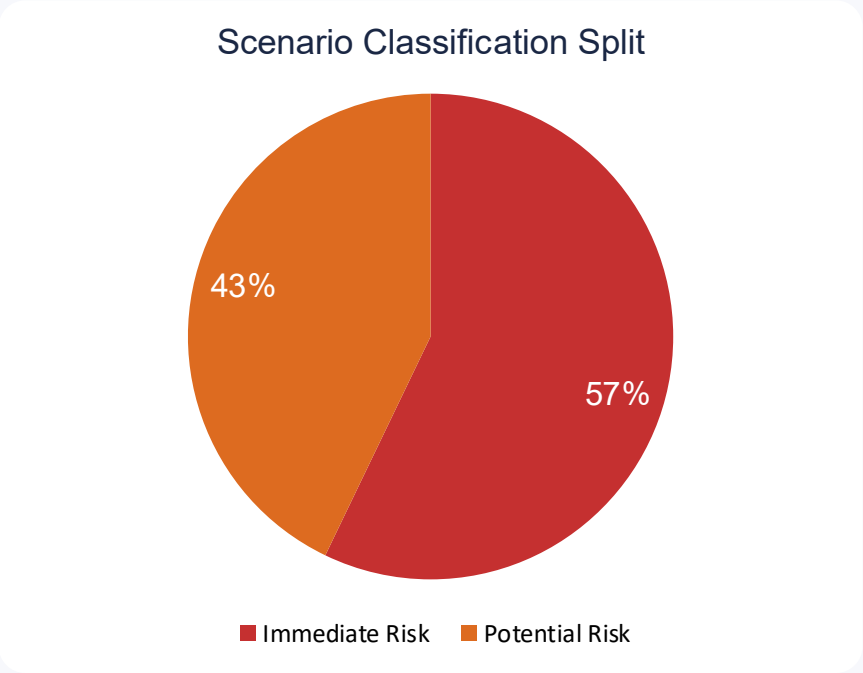
07 Elevated Latency — Within SLA

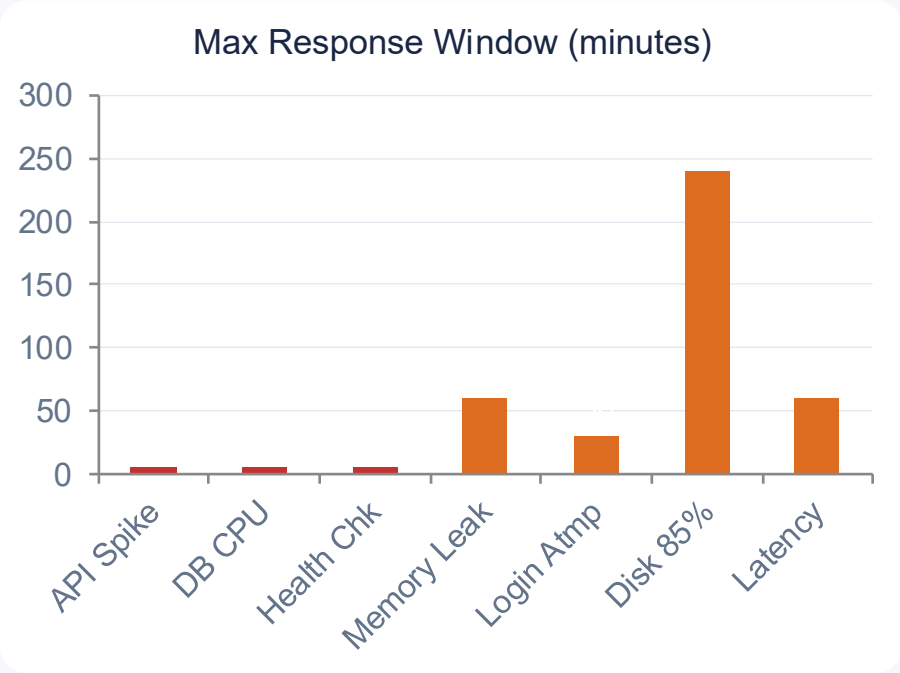
POTENTIAL

Alert: Latency increased 80ms → 180ms; SLA threshold is 300ms

Reason: No SLA breach yet; could indicate degradation trend; still within acceptable bounds **Response:** Monitor and investigate; optimize before threshold is crossed

Classification Distribution & Response Times





Red = Immediate Risk (0–5 min) Orange = Potential Risk (varies)

QUICK DECISION FRAMEWORK

3-Step Triage Process

1 Is impact happening NOW? → YES → IMMEDIATE RISK

NO →

2 Will impact occur within minutes? → YES → IMMEDIATE RISK

NO →

3 Is this a trend or warning signal? → YES → POTENTIAL RISK

Rule of Thumb: "Are users or systems being impacted right now, or within the next few minutes?" YES = Immediate Risk | NO = Potential Risk

COMPARISON SUMMARY

Immediate vs. Potential Risk at a Glance

Dimension	IMMEDIATE RISK	POTENTIAL RISK
Timeframe	Now or within minutes	Hours to days
Current Impact	Active or imminent	None yet
Urgency	Interrupt-driven (page)	Scheduled response
Primary Focus	Mitigation & stabilization	Prevention & investigation
Tolerance for Delay	Low — minutes matter	Moderate — planned response
Response SLA	0–5 minutes	15–60 minutes

KEY TAKEAWAYS

What Every Responder Must Remember

Speed Matters for Immediate Risks

If impact is active or minutes away, every second counts. Begin mitigation before root cause is fully understood. Stabilization first.

Don't Ignore Potential Risks

Early warning signals prevent incidents. Potential risks left unaddressed become immediate risks. Track and remediate proactively.

Use the 3-Step Triage Framework

Ask: (1) Impact now? (2) Within minutes? (3) Trend or warning? Consistent classification prevents under- and over-reaction.

Context Defines Blast Radius

A single-region failure may be Immediate Risk when redundancy is being consumed. Always evaluate the broader system impact.

Track Trends, Not Just Thresholds

Gradual degradation (memory, disk, latency) are Potential Risks with predictable timelines. Monitor trends, not just breach events.

Document & Escalate Appropriately

Immediate Risks: page and escalate. Potential Risks: open a ticket and schedule. Proper documentation prevents repeated incidents.